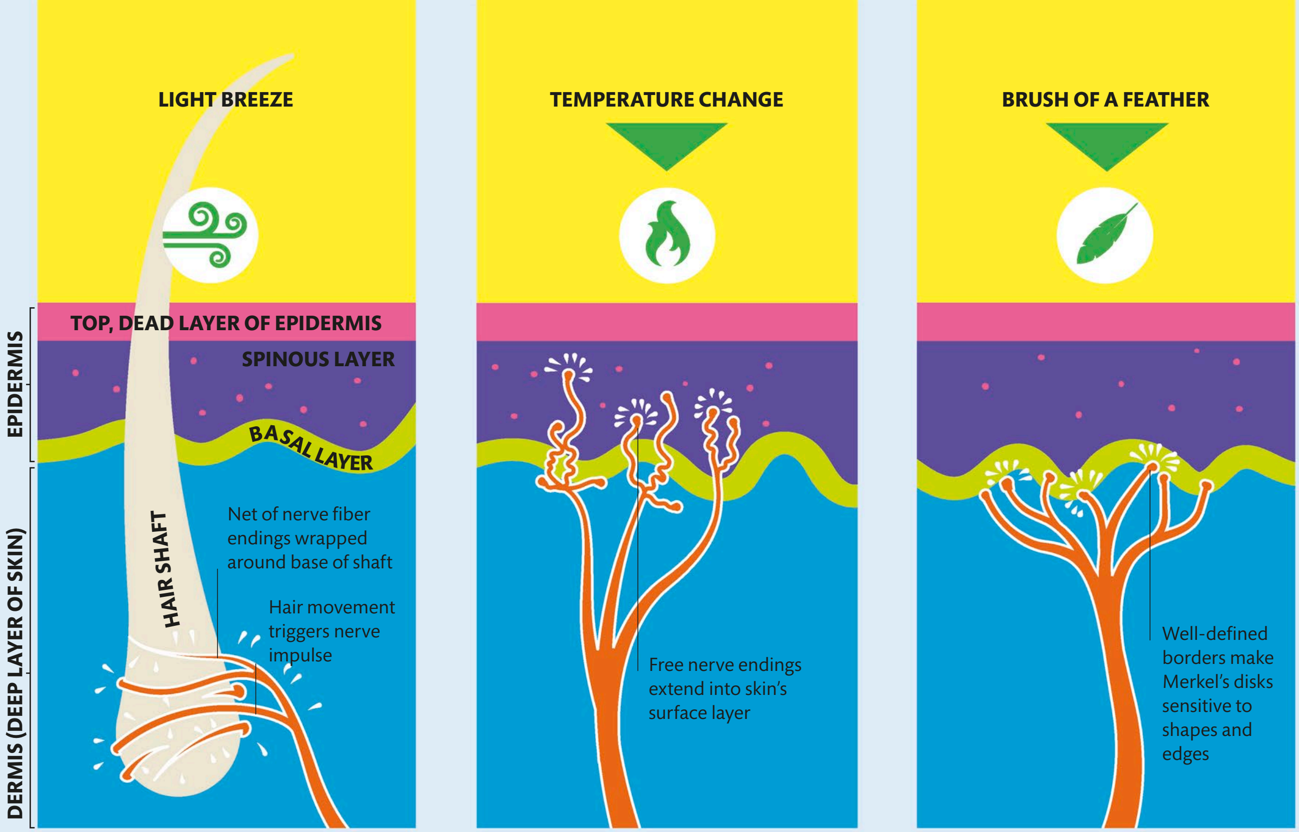

Root hair plexus
Nerves wrapped around the base of a hair shaft are triggered by things that have not touched the skin, such as air currents or objects that brush against the hair.

Free nerve endings
Extending up into the spinous layer of the epidermis, these bare, rootlike nerve endings are sensitive to cold, heat, light touch, and pain.

Merkel's disks
Found slightly lower than free nerve endings, Merkel's disks are particularly dense in the lips and fingertips. They respond to light touch.

Touch

The skin is the biggest organ of the body and also the largest sense organ. Packed with sensors, it enables us to experience a wide variety of sensations, as well as an awareness of where we are.

Receptors in the skin

Skin sensors consist of receptors connected by axons. Found at various levels in the skin, there are around 20 types that respond to different sorts of stimuli. The receptors register mechanical, thermal, and, in some cases, chemical stimuli and convert them into electrical signals. These travel up peripheral nerves to the spinal cord, then to the brain stem, and finally to the somatosensory cortex, where they are translated into a touch.

TYPES OF RECEPTORS	FUNCTION
Mechanoreceptors	Sensory receptors that respond to mechanical pressure or distortion. This can range from a light touch to deep pressure.
Proprioceptors	Receptors that receive stimuli from within the body, particularly in relation to position and movement.
Nociceptors	Sensory neurons that respond to damaging stimuli by sending "possible threat" signals to the spinal cord and the brain.
Thermoreceptors	Specialized nerve cells that are able to detect differences in temperature. They are found all over the skin and in some internal areas.
Chemoreceptors	Extensions of the peripheral nervous system that respond to changes in blood concentrations to maintain homeostasis (see pp.90–91).